TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.6 ± 1.62
Correlation Stability	84.7 ± 0.90
Distribution Consistency	86.5 ± 2.20
Entropy Preservation	90.7 ± 1.55
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.8 ± 0.40
Feature Accuracy	95.9 ± 0.94
Novelty Score	0.9 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	40.629	-0.571	1.4%	OK
Sex	1.103	1.105	+0.002	0.1%	OK
Steroid	1.506	1.605	+0.098	6.5%	OK
Antivirals	1.845	1.879	+0.034	1.8%	OK
Fatigue	1.351	1.411	+0.061	4.5%	OK
Malaise	1.604	1.661	+0.057	3.6%	OK
Anorexia	1.792	1.798	+0.006	0.3%	OK
Liver Big	1.828	1.855	+0.027	1.5%	OK
Liver Firm	1.583	1.661	+0.078	4.9%	OK
Spleen Palpable	1.800	1.782	-0.018	1.0%	OK
Spiders	1.660	1.661	+0.001	0.1%	OK
Ascites	1.867	1.879	+0.012	0.7%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.6 ± 1.62
Correlation Stability (*2)	84.7 ± 0.90
Distribution Consistency (*3)	86.5 ± 2.20
Entropy Preservation (*4)	90.7 ± 1.55
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.8 ± 0.40
Feature Accuracy (*10)	95.9 ± 0.94
Novelty Score (*11)	0.9 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.867741935483871	0.8064516129032258	0.8258064516129032	-0.04193548387096775	
F1	0.5522765469824293	0.4011655011655012	0.5028787878787879	-0.04939775910364144	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.7903225806451613	0.8129032258064516	-0.03870967741935485	
F1	0.5385593354014405	0.4081709956709957	0.5049728049728055	-0.03358653042863571	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7774193548387097	0.8193548387096774	0.02903225806451613	
F1	0.561449871078354	0.5084365926471189	0.593403720462544	0.03195384938418995	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7935483870967742	0.7709677419354839	0.7709677419354839	-0.02258064516129032	
F1	0.4674278499278497	0.4313752913752914	0.4259407749113631	-0.04148707501648676	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8419354838709676	0.7322580645161291	0.8032258064516128	-0.03870967741935481	
F1	0.5331626706626706	0.2529534844240726	0.4307575757575758	-0.1024050949050949	

Model: LR

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.7741935483870968	0.8290322580645161	-0.009677419354838724	
F1	0.5132478632478632	0.2819741369741371	0.4866142191142191	-0.026633644133644133	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8548387096774193	0.7741935483870968	0.8354838709677418	-0.019354838709677413	
F1	0.5837338100495995	0.3810866910866911	0.5424908424908425	-0.041242967558757025	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.7935483870967742	0.7806451612903226	0	
F1	N/A	0.045	N/A	0	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.8064516129032258	0.8064516129032258	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.7612903225806452	0.8096774193548386	-0.029032258064516148	
F1	0.5718392134181608	0.3681929181929181	0.4433080808080808	-0.12853113261007998	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	24.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
2	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	67.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	47.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
5	36.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
6	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
2	20.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	1.0
3	48.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
4	37.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
5	39.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
6	36.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	51.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
2	50.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
3	36.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	36.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
5	48.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	51.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	59.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
2	28.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
3	61.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	27.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	42.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	28.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
2	22.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0
3	48.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	44.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
5	48.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	27.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	62.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	1.0	2.0
2	58.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
3	67.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	42.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	37.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
6	37.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	51.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	45.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	56.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	56.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	47.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
6	28.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	54.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
5	62.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
6	31.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	61.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
2	51.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	25.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	1.0
4	28.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0	2.0
5	49.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
6	54.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	32.0	2.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	31.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	43.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
4	42.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
5	34.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0